In Silico Screening of BCL-2 Inhibitors Using Machine Learning, Molecular Docking, and Molecular Dynamics Approaches

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***Abstract*—In the fight against cancer, developing effective Bcl-2 inhibitors is crucial for enhancing apoptosis in cancer cells. This study leverages machine learning models to identify potential inhibitors using open data sources from the PubChem and ChEMBL databases. These compounds were marked as 'active' or 'inactive' based on their Bcl-2 inhibitory potential. Using the Random Forest model with compound SMILES encrypted in Morgan3 Fingerprint, 17 were identified as 'active' and met ADMET criteria from more than 6,000 screened compounds. These compounds will undergo molecular docking to predict binding affinity, followed by molecular dynamics (MD) simulations to assess stability, flexibility, and interactions with Bcl-2 over time. MD simulations will provide insights into conformational changes and binding persistence, aiding in the selection of promising candidates. This research aims to identify novel Bcl-2 inhibitors with favorable ADMET profiles for targeted cancer therapy. The integration of computational techniques, including machine learning, molecular docking, and MD simulations, enhances the drug discovery process, accelerating the identification of potential candidates for preclinical and clinical development.**

Keywords—Bcl-2 inhibitor, in silico screening, Random Forest model, molecular docking, molecular dynamics

I. Introduction

Cancer remains a critical global health issue, with nearly 20 million new cases and 9.7 million deaths recorded in 2022 [1]. The most common causes of cancer deaths are lung, liver, and stomach cancers. Asia bears the largest burden, accounting for half of these cases. In Vietnam alone, over 120,000 cancer-related deaths were reported in 2022, with liver, lung, and stomach cancers being the most prevalent [2]. The estimated 16% lifetime risk of developing cancer in Vietnam underscores the need for more effective diagnostic and therapeutic strategies.

Cancer is a heterogeneous group of diseases marked by uncontrolled cell proliferation and the potential to invade surrounding tissues and metastasize. Metastasis, responsible for over 90% of cancer deaths, occurs when malignant cells disseminate via blood or lymphatic systems to distant organs [3]. The mechanisms of cancer development, or carcinogenesis, is driven by complex interactions between genetic mutations and environmental exposures. Activation of oncogenes, inactivation of tumor suppressor genes, and defects in DNA repair systems contribute to uncontrolled growth [4]. External factors such as tobacco smoke and industrial carcinogens further accelerate genetic damage [5]. A key consequence of these genetic and environmental insults is the disruption of apoptosis, the programmed cell death mechanism that ordinarily eliminates damaged or abnormal cells [6].

Apoptosis, or programmed cell death, is vital for cellular homeostasis and development. It proceeds through two main pathways: the intrinsic (mitochondrial) pathway and the extrinsic (death receptor) pathway, both converging on the activation of caspases that dismantle the cell [7]. In cancer, apoptotic pathways are often disrupted, allowing abnormal cells to persist and proliferate. The Bcl-2 protein family plays a central role in regulating intrinsic apoptosis. It includes both pro-apoptotic (e.g., Bax, Bak, BH3-only proteins) and anti-apoptotic (e.g., Bcl-2, Bcl-xL, Mcl-1) members that regulate mitochondrial outer membrane permeabilization (MOMP), a decisive step in cell death [8], [9]. These proteins share conserved Bcl-2 homology (BH) domains, which are crucial for their interactions and functions.

Targeting apoptosis has emerged as a promising therapeutic strategy. Drugs such as venetoclax, a selective Bcl-2 inhibitor, have demonstrated clinical efficacy by restoring apoptosis in cancer cells [10]. Other novel agents, targeting oncogenic signaling pathways mentioned include MEK inhibitors like selumetinib and trametinib, BH3 mimetics such as navitoclax and venetoclax, and specific KRAS-G12C inhibitors like AMG 510 and MRTX849. Additionally, TRK inhibitors like larotrectinib and entrectinib, along with proteasome inhibitors, have been developed to disrupt these pathways and promote apoptosis in cancer cells [11].

Despite those progresses, integration of machine learning (ML) into Bcl-2 inhibitor discovery remains limited, and research often targets only specific cancer types. This study addresses these gaps by implementing ML algorithms such as support vector machines and random forests for virtual

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screening to identify and validate novel Bcl-2 inhibitors. It also evaluates drug-likeness and efficacy across multiple cancers, especially those with high incidence in Vietnam. By integrating ML and molecular docking, the study builds a robust in silico pipeline to accelerate drug discovery. This approach enhances therapeutic precision and supports the development of personalized cancer treatments through the fusion of computational methods and molecular targeting..

II. METHODS

A. Data Acquisition

The compounds in the training data set were obtained from the ChEMBL public database using Bcl-2 UniProt ID of P10415. Then, the retrieved data was cleaned by deleting null data and dropping duplicated data. Molecular fingerprints were generated from the compound SMILES using 166 bit Molecular Access System (MACCS) and 1024 bit Morgan3 fingerprints encoder. Compounds with half-maximal inhibitory concentration values $IC_{50} \leqslant 0.5\ \mu M$ were defined as "active" samples, and those with higher values were defined as "inactive" ones.

The screening data set was constructed using gene ID '596' from PubChem, To ensure data relevance, only assays labeled as 'Confirmatory' and with a 'PUBCHEM_ACTIVITY_OUTCOME' of 'Activity' were retained. Lipinski descriptors were calculated using RDKit and saved in CSV format. PubChem fingerprints were generated from SMILES using two mentioned fingerprints above.

B. ML model optimization and testing

To build the expected model of the compound's Bcl2 inhibitory activity, different types of models were used including Support Vector Machine (SVM) and Random Forest (RF). These models were employed in Python language on the Google Colaboratory platform. The model hyperparameters tuning was conducted using a grid search technique with 5-fold cross-validation through the Python scikit-learn package. The data retrieved from the ChEMBL public database was divided into a training set (80%) and a validation set (20%). The best parameters were saved based on the accuracy value from the cross-validation used inside the function. Next, the model with the best performance was applied to the screening dataset achieved from PubChem to predict the potential "active" compounds. The performance results, including accuracy, precision, recall, F1 score, and AUC, were calculated.

C. Molecular dockings

After identifying compounds with potential to inhibit Bcl2 using the ML model with the best performance, molecular docking was conducted to analyze the chemical interactions between Bcl2 active site and the ML-screened compounds. The 3D structure of Bcl2 was collected from the RCSB Protein Data Bank database (PDB ID: 6O0K) [12]. Prior to the docking, all 3D structures of protein and compounds were prepared by removing water molecules, adding missing hydrogen atoms and assigning charges using the Dock Prep tool inside UCSF Chimera software with version 1.17.3. The docking box dimension is 26 Å × 26 Å × 26 Å and its center coordinates are x = − 14.226, y = 1.146 and z = − 10.800 [13].

Next, the docking process was performed using AutoDock Vina 1.1.2 plugin to obtain the optimal receptor–ligand binding poses and the correspond docking score or the binding energy.

Finally, Biovia Discovery Studio 22.1 was used to visualize binding modes and export the 2D interaction maps between Bcl2 and compounds.

D. Molecular dynamics (MD) simulations

The Bcl2 apo-protein and three complexs with best docking scores were further subjected to MD simulations using Gromacs 2024.1 [14]. Topology files were generated using the CHARMM36 force field, and the TIP3P water model, as recommended by GROMACS, was employed. Ligand topology files were prepared using the CGENFF web server to generate ".str" files, which were manually integrated into the complex topologies via custom Python scripts [15].

Each protein–ligand complex was placed in a dodecahedral box with a minimum distance of 1 nm between the solute and the box wall. The systems were solvated using the SPC216 explicit water model, ionized, and neutralized. Energy minimization was performed using the steepest descent algorithm for up to 500 ps, or until convergence to a stable conformation was achieved. Systems were then gradually heated from 0 K to 300 K.

Equilibration was conducted in two phases with a 2 fs time step under position restraints. The first phase involved a 1 ns simulation in the NVT ensemble using the modified Berendsen thermostat at 300 K. This was followed by a 1 ns NPT equilibration using the Parrinello–Rahman barostat at 1 bar pressure. Long-range electrostatic interactions were treated with the Particle Mesh Ewald (PME) method, and a 1 nm cutoff was applied for short-range electrostatic and van der Waals interactions. Hydrogen bonds were constrained throughout using the LINCS algorithm.

Finally, a 100 ns production run was performed for each system, with trajectory snapshots recorded every 10 ps for subsequent analysis [16].

E. In silico ADME and toxicity analysis

Compounds with top-ranking binding energy scores were subjected to in silico ADME prediction using the SwissADME web tool (www.swissadme.ch). The SMILES format of each compound was used to assess ADME-related properties, including molecular weight, hydrogen bond acceptors/donors, topological polar surface area (TPSA), lipophilicity (log P), water solubility (log S), blood–brain barrier (BBB) permeability, skin permeability (log Kp), pharmacokinetics, and drug-likeness. In silico toxicity was predicted using the ProTox 3.0 web tool (https://tox.charite.de/protox3/), with evaluations based on predicted toxicity class and median lethal dose (LD_{50}) [17].

III. RESULTS AND DISCUSSIONS

A. Data acquisiton

For the training set, 836 input vectors were used to build different ML models with 599 compounds classified into the "active" group and 237 compounds classified into the "inactive" group. Regarding the screening set, we collected 6025 small molecules from PubChem data. A Tanimoto score analysis was performed to measure the similarity between two datasets as shown in Fig. 1. The average Tanimoto score

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between these two files is 0.1664. Moreover, the score distribution peaks predominantly within the 0.1–0.3 range, indicating low similarity among most compound pairs. This suggests that the dataset is sufficiently diverse, making it well-suited for machine learning prediction modeling, as such diversity enhances the model's ability to generalize to novel compounds.

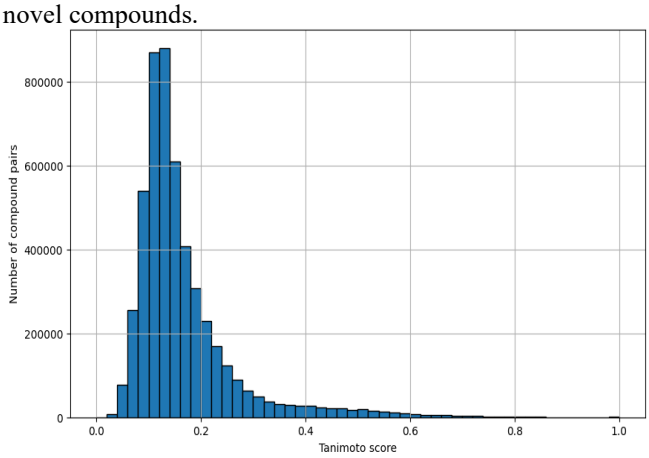


Fig. 1. Distribution of Tanimoto scores between two data sets.

B. Machine learning model performance and results

The optimal hyperparameter values and performance metrics for each model, using different molecular descriptors, are summarized in Table 1. Among the evaluated models, the Random Forest (RF) model employing the Morgan3 fingerprint demonstrated the best performance, achieving an accuracy of 0.85, sensitivity of 0.88, specificity of 0.78, and an area under the ROC curve (AUC) of 0.92. This model was subsequently used to screen a set of candidate compounds from the PubChem database.

A total of 6,025 primary compounds were analyzed, of which 3,550 were predicted as "active" and 2,475 as "inactive." Applying Lipinski's Rule of Five (Ro5), only 17 of the predicted active compounds met the Ro5 criteria (Table 2), while 3,533 did not. Among the inactive compounds, 1,035 satisfied the Ro5 criteria, whereas 1,440 did not.

TABLE 1. THE OPTIMAL HYPER-PARAMETER VALUES OF EACH MODEL AND THEIR CORRESPONDING PERFORMANCES.

Models	Hyperparameters	Performance
SVM-Morgan3	Kernel = rbf Gamma = 10 C = 10	Accuracy: 0.85 Sensitivity: 0.88 Specificity: 0.77 AUC: 0.90
SVM-MACCS	Kernel = poly C = 10 Gamma = 0.01	Accuracy: 0.86 Sensitivity: 0.90 Specificity: 0.77 AUC: 0.90
RF-MACCS	n_estimators =150 criterion = entropy max_depth = None min_samples_split = 2 min_samples_leaf = 1	Accuracy: 0.85 Sensitivity: 0.90 Specificity: 0.73 AUC: 0.92
RF-Morgan 3	**n_estimators = 50 criterion = entropy max_depth = 10 min_samples_split = 2 min_samples_leaf = 4**	**Accuracy: 0.85 Sensitivity: 0.88 Specificity: 0.78 AUC: 0.92**

C. Molecular docking results

The molecular docking results of 17 active compounds are summarized in Table 2. The ligands bonded with Bcl2 that exhibited the highest docking scores were 70695485 (−9.0 kcal/mol), 70697508 (−8.9 kcal/mol), and 90083384 (−8.7 kcal/mol) and their 2D interaction maps were shown in Fig. 2. All compounds interacted with the amino acids of Bcl2 through Van der Waals interaction, hydrogen bonds, π-alkyl bonds, alkyl bonds, π- π stacking, π- σ bonds and electrostatic interactions. These results are consistent with other studies that docked Venetoclax with Bcl2 binding site. These results emphasized the potential of the RF model to screen compounds for Bcl2 inbibitors.

TABLE 2. MOLECULAR DOCKING RESULTS OF COMPOUNDS WITH BCL2

Compound ID	Docking Score (kcal/mol)	Amino Acid Residues
70695485	-9.0	Phe104, Asp111, Phe112, Met115, Leu137, Ala149, Val156, Tyr202
70697508	-8.9	Phe104, Phe112, Asp111, Met115, Leu137, Ala149, Val156, Tyr202
90083384	-8.7	Phe104, Tyr108, Asp111, Met115, Val133,Glu136, Leu137,Val156
5926970	-8.3	Phe104, Tyr108, Asp111, Met115, Ala149
12006175	-8.3	Phe104, Phe112, Arg146, Ala149, Val156
1561895	-7.9	Phe104, Tyr108, Met115, Leu137, Ala149, Val156
5046669	-7.8	Phe104, Asp111, Glu136, Arg146
654918	-7.8	Phe104, Asp111, Leu137, Asn143, Ala149
118535457	-7.7	Phe104, Tyr108, Met115, Ala149, Val156
5051642	-7.3	Phe112, Met115, Leu137, Ala149
4173573	-7.3	Tyr108, Met115, Leu137
1919075	-6.9	Phe104, Tyr108,Asp111, Met115,Val133, Arg146, Ala149
11502754	-6.9	Asp111, Met115, Leu137, Arg146, Ala149, Val156
117937598	-6.7	Tyr108, Trp144, Gly145, Tyr202
117937605	-6.7	Ala100, Trp144, Gly145, Tyr202
90082812	-6.5	Asp111, Met115, Val133, Leu137, Arg146, Ala149
2043339	-6.3	Phe104, Tyr108, Phe112, Met115, Val133, Leu137, Ala149, Phe153, Val156

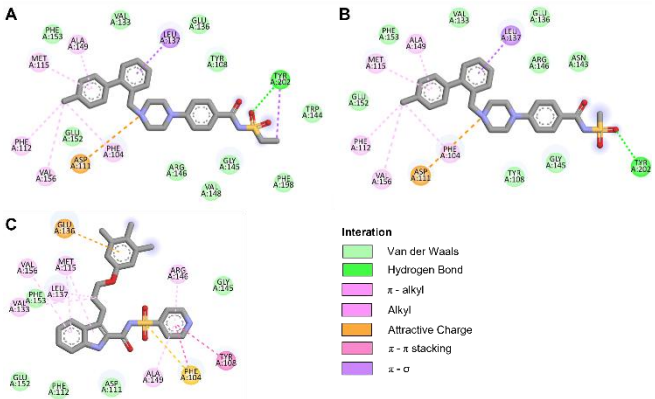


Fig. 2. 2D interaction maps between three compounds with the highest docking scores and BCl-2: (A) Bcl2/70695485, (B) Bcl2/70697508 (green), and (C) Bcl2/90083384.

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D. Molecular dynamic results

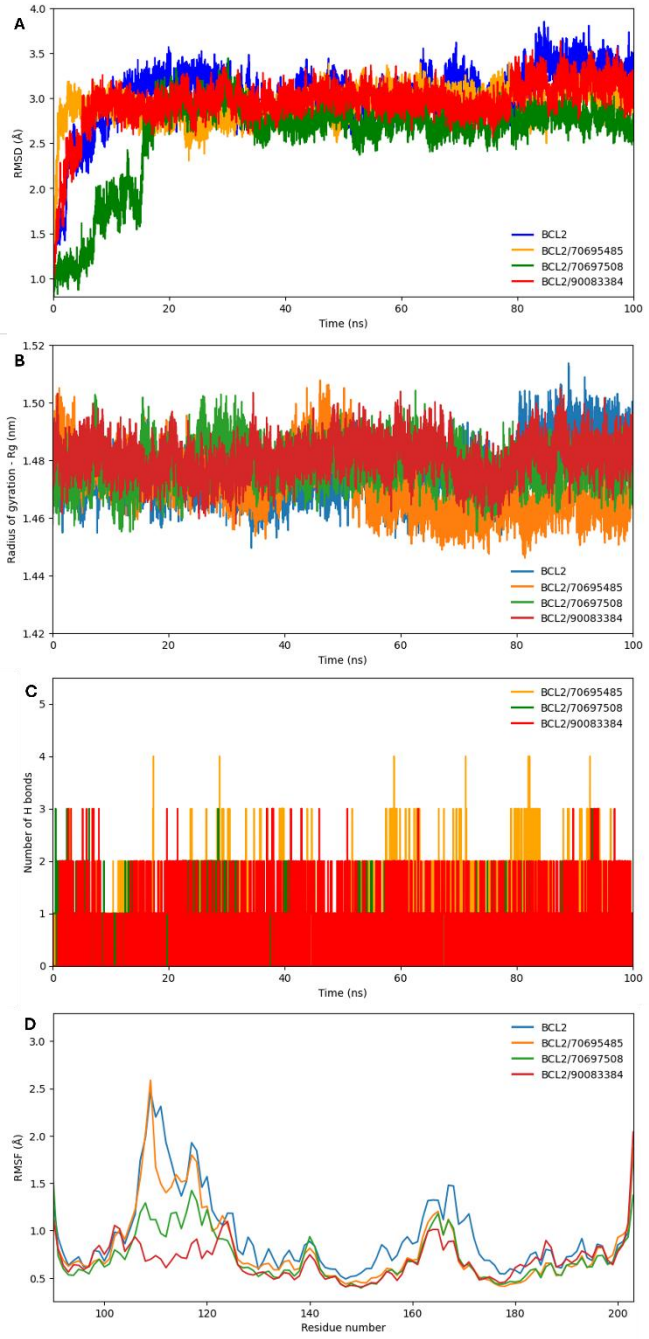


Fig. 3. MD simulation analysis of Bcl2 (blue), Bcl2/70695485 (orange), Bcl2/70697508 (green), and Bcl2/90083384 (red): (A) Root mean square deviation, (B) Radius of gyration, (C) Hydrogen bond number, and (D) Root mean square fluctuation.

Three selected compounds with the best docking scores when binding to Bcl2 including 70695485, 70697508, and 90083384 are further evaluated through MD simulation analysis in 100 ns. Except the initial equilibration time of almost 20 ns, the RMSD values fluctuated over a moderate range of approximately 2.5–3.5 Å till the end of the simulation as illustrated in Fig. 3A. More importantly, the deviation values of Bcl2 were higher than those of the three complexes throughout the simulation period, which signifies favorable binding of the three selected compounds to the Bcl2 target. In particular, the RMSD value of Bcl2 was in the range of 0.7–3.0 Å in the first 20 ns and stabilized at 3.0–3.6 Å in the next 80 ns. Meanwhile, the RMSD of Bcl2/70695485 and Bcl2/90083384 complexes are in range of 2.5–3.5 Å and higher than that of Bcl2/70697508.

The radius of gyration (Rg) values of Bcl2 and three complexes as shown in Fig. 3B show only minor changes in the entire simulation time, indicating that their structures were minimally affected. In consistency with the RMSD results, the Rg values of the apoprotein Bcl2, Bcl2/70695485, and Bcl2/90083384 Bcl2/70697508 were 1.4755± 0.0089 nm, 1.4805 ± 0.0065nm, 1.4714 ± 0.0100nm and 1.4779 ± 0.0067nm, respectively. There is no significant change in Rg values for all three complexes, therefore, the protein structures with all three ligands revealed overall stable protein folding without any significant expansion or condensation

The number of hydrogen bonds formed between three compounds and the Bcl2 binding site fluctuated between 1 and 4 during 100 ns of simulation, as shown in Fig. 3C. Notably, compound 70695485 established the highest number of hydrogen bonds (1–4 bonds) with Bcl2, followed by compound 90083384 with 1–3 bonds and then compound 70697508 with 1–2 bonds.

Figure 3D presents the Root Mean Square Fluctuation (RMSF) values for the Bcl-2 apo-protein and its complexes with three selected ligands. Overall, the RMSF values for the apo-protein are higher than those of the ligand-bound complexes, particularly in the regions spanning residues 110–138 and 142–179. This suggests that ligand binding reduces the flexibility of these residues, indicating successful interaction and stabilization of the protein structure. Across all complexes, RMSF values range between 0.5 and 2.6 Å, reflecting moderate atomic fluctuations upon ligand binding.

E. ADMET Evaluation

All compounds demonstrated high gastrointestinal absorption, indicating their potential suitability for oral pharmaceutical applications. However, none were able to cross the blood–brain barrier, which may limit their use in treating central nervous system (CNS)-related disorders. The Log Kp values ranged from −7.34 to −5.53 cm/s, reflecting low skin permeability and suggesting limited potential for transdermal delivery.

In terms of cytochrome P450 (CYP) enzyme interactions, all compounds exhibited varying degrees of inhibition against key isoforms, including CYP1A2, CYP2C19, CYP2C9, CYP2D6, and CYP3A4. Notably, compound 70697508 inhibited all tested CYP enzymes except CYP2D6, indicating a high likelihood of drug–drug interactions and necessitating caution in combination therapies.

Among the evaluated compounds, 70695485 emerged as the most promising candidate. It exhibited strong gastrointestinal absorption, relatively higher skin permeability compared to other compounds, and inhibited multiple CYP enzymes (CYP1A2, CYP2C19, CYP2C9, and CYP3A4) but spared CYP2D6. This profile reduces the potential for adverse drug interactions while maintaining therapeutic efficacy across various applications. Although its inability to penetrate the blood–brain barrier remains a limitation, it may still represent a strong candidate for further development targeting non-CNS-related conditions. The ADME results are summarised in Table 3.

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TABLE 3. ADME ANALYSIS OF TOP COMPOUNDS

Compound	LogKp (cm/s)	GI Absorption	BBB permeant	P-glycoprotein substrate	CYP1A2 inhibitor	CYP2C19 inhibitor	CYP2C9 inhibitor	CYP2D6 inhibitor	CYP3A4 inhibitor
70695485	-5.92	High	No	No	No	Yes	Yes	Yes	Yes
70697508	-6.09	High	No	No	No	Yes	Yes	Yes	Yes
90083384	-5.53	Low	No	No	No	Yes	Yes	Yes	Yes

IV. CONCLUSION

This study successfully identified 17 potential Bcl-2 inhibitors using a combination of machine learning and molecular docking. These compounds exhibited strong binding affinities and favorable interactions with the Bcl-2 protein, indicating their promise as candidate anticancer agents. The application of advanced in silico methods enabled the efficient screening of thousands of compounds, underscoring the utility of computational approaches in modern drug discovery. These findings contribute to ongoing efforts to develop targeted cancer therapies by providing insight into molecular mechanisms underlying Bcl-2 inhibition. The integration of computational techniques with traditional drug development pipelines may accelerate the discovery of more effective and personalized treatments. While the identified compounds show significant potential, further validation through experimental studies is essential to assess their efficacy and safety. Continued refinement of computational models, in tandem with laboratory-based research, will be key to translating these discoveries into clinically viable therapies, advancing the goal of precision oncology.

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